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A cat is not a battleship: thoughts on the meaning of "neuropsychoanalysis"

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ABSTRACT
As a conspicuously hybrid entity, neuropsychoanalysis enjoins one to look critically at its assumptions about knowledge and subjectivity as one tries to understand how its un-hyphenated halves relate to one another. The author looks at the differences between mind (which is grounded in subjective experience) and brain (which is an objectively described neurobiological entity), and suggests that neuropsychoanalytic writers are inclined to acknowledge but then disregard the unique, irreducible nature of lived experience, and the fundamental differences between the psychoanalytic mind (which requires an experiencing subject) and the brain (which is a neuronal aggregate). The author offers a philosophical basis for contending that there are potential dangers for psychoanalysis when neuroscience is misrecognized in its fundamental differences and injudiciously employed as a psychoanalytic partner in order to answer questions that properly belong to the language and conceptual architecture of psychoanalysis.

KEYWORDS
Hermeneutic; scientism; epistemology; subjectivity; neuropsychoanalysis; phenomenology

> [T]he theory and practice of psychoanalysis presume or induce an attitude of mind and even, all unknown to Freud, a "new philosophy" … Doctor Hesnard approves of those who take care … to separate psychoanalysis from a scientistic or objectivist ideology … Here phenomenology brings to psychoanalysis certain categories, certain means of expression that it needs in order to be completely itself. (Merleau-Ponty 1960, 67)

Eric Kandel (1998, 2005), the Nobel Prize-winning proponent of a neurobiological reformation in psychoanalysis, has remarked that the great error in the development of psychoanalysis came when it did not align itself from the beginning with neurobiological science. "Because psychoanalysis has not yet recognized itself as a branch of biology," he writes, "it has not incorporated into the psychoanalytic view of the mind the rich harvest of knowledge about the biology of the brain and its control of behaviour as it has emerged in the last 50 years" (Kandel 2005, 67). In this view, only by turning decisively to neurobiology—and reconfiguring Freudian psychoanalysis to fit the findings of brain science—can psychoanalysis rescue itself from obsolescence.

The corrective promise of neuropsychoanalysis almost always includes a renewal of Freud's original scientific program. Fonagy's (2003) assertion that only by way of systematic research will psychoanalysis ever sit at the "high table of the scientific study of the mind" (p. 232) is echoed by Turnbull and Solms (2003) with respect to neuroscience:

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"The high road for psychoanalysis is to engage with the neuroscientific issues which should now directly interest it. This will not be an easy task … . An increasing number of psychoanalysts today are, however, keen to rise to the challenge … " Also, we are told, should the *cris de cœur* be heard and responded to, then "A radically different psychoanalysis will emerge" (p. 82). This is no doubt true. The question, of course, is whether or not psychoanalysis as we know it can stand up to the philosophical complications that neuropsychoanalysis brings to the field, keeping in mind that philosophical confusion can lead to real consequences on the ground, because such confusion will inevitably seep into how we think (theory) and what we choose to do (practice).

When neuropsychoanalytic authors address problems of epistemological contradiction between psychoanalysis and neuroscience, they usually do so in one of two ways: either a common "objectivist" epistemology is argued or assumed, making psychoanalytic knowledge continuous with knowledge from other sciences, or a pluralism of epistemic approaches is acknowledged but not acknowledged as problematic. Neither response addresses the question of how psychoanalysis, once it enters into partnership with neuroscience, retains its basic identity as the "idiosyncratic science of the individual subject" (Caws 2003). As an explicitly hybrid entity, neuropsychoanalysis *by its very nature* enjoins us to begin with questions about how, for example, a poetics of dreaming (guided by the subjective experience of dreaming itself, given form by a relational process of narrating and listening, and organized according to psychoanalytic rules of interpretation) should engage with neurocognitive memory studies or neuro-imaging studies of the dreaming brain, whose interests and organizing assumptions are so fundamentally different. I begin from the premise that, until these questions are addressed, the ostensive hybridity of neuropsychoanalysis risks being no more than a juxtaposition of unrelated terms.

The evolution of the neuropsychoanalytic movement has been rapid and impressively influential, perhaps especially in the Anglo-American world. Since the early 1990s many have championed a neurobiological reform of psychoanalysis, among them Yovell, Solms, and Fotopoulou (2015); Kandel (1998, 2005), Solms (1998), Kaplan-Solms and Solms (2000), Pally (2000), Solms and Turnbull (2002), Opatow (1999), and Turnbull and Solms (2003).[1] This position has drawn energy from a rank-and-file of practicing analysts, many of whom have felt dispirited by the welter of theories they confronted in their training or by the growing disrespect or disinterest of professional colleagues and competitors, and who perhaps were drawn to the appealing clarities of empirical research and neurobiological science.

Several authors, however, have critically addressed the pull towards neuroscience. Blass and Carmeli (2015, 2007) detail what they regard as the endemic fallacies in neuropsychoanalytic writing. Brothers (2002) points out the significant limitations and irresponsible speculations in "neuroist" positions, as well as the misunderstandings of basic research on the part of some the authors of those positions. Green (1999, 2000, 2001, 2002, 2005a, 2005b, 2008) has cast suspicion on neuropsychoanalytic tenets in more directly epistemological ways. Karlsson (2010), drawing from phenomenology, has addressed Solms and Turnbull's (2002) adoption of a "dual-aspect monist" perspective on the mind–body

[1]Others include Modell (1993, 2003, 2012a, 2012b, 2013a, 2013b), Olds (2012), Olds and Cooper (1997), Gabbard (2000a, 2000b), Westen and Gabbard (2002a, 2002b), Cooper (1993), Schore (1994, 2003, 2012), Green (2003), Fotopoulou (2012a, 2012b, 2012c), Shevrin (1999a, 1999b, 2012), Pulver (2003), Damasio (2010, 2012), Zellner (2012), and Ginot (2015).

problem as a ground for their elaboration of a neuropsychoanalytic position, and uncovered both contradictions in their argument as well as a disguised "naturalistic" or scientific epistemology. Hoffman (2009) has argued against the "desiccation" of the human in psychoanalytic discourse due to the imposition of systematic research, both as an epistemological and a political force. Orange (2001, 2003) has explored philosophical issues that bear directly on problems of psychoanalytic epistemology. The philosopher Talia Morag (2016), taking an epistemological position she and her colleague David Armstrong call "liberal naturalism," also clears a path for a non-scientific (and non-neuropsychoanalytic) psychoanalysis. In two books, the neuroscientist M.R. Bennett and philosopher P.M.S. Hacker (2003, 2007) mount carefully reasoned arguments against the philosophical ideas that underwrite neuroscience and (by extension) neuropsychoanalysis. Ricoeur's (1970) seminal philosophical study of Freud appropriates instinct (biology) as a discursive mental phenomenon, and characterizes psychoanalysis as a "hermeneutics of suspicion," a hybrid discourse combining dialectically a language of force or energy (causes) and a language of meaning (reasons and motivations).

Blass and Carmeli's critique of neuropsychoanalysis

In some respects, one would be hard pressed to improve upon Blass and Carmeli's critique of neuropsychoanalysis. They focus with admirable concision on the key areas of neuropsychoanalytic confusion and (at times) misrepresentation. Their 2015 critique is organized around what they see as the four main arguments put forward by Yovell, Solms, and Fotopoulou (2015) in their effort to make the positive case for neuropsychoanalysis. Blass and Carmeli's (2015) summary and response to each argument follows.

(1) Yovell et al. argue: Psychoanalysis and neuroscience study the same entity, the so-called "mindbrain," and therefore knowledge of one contributes to the other. They are the same entity due to the dependent relationship between the mental and its biological substrate.

Blass and Carmeli respond, "the analyst concerned with a meaningful process of understanding the patient's inner psychical dynamics has no [more] need to be concerned with the biological matter of the brain" than the painter with the chemical or molecular structure of his paint (pp. 1560–1561). It is not, in other words, relevant to the analyst's *psychoanalytic* purpose, because the two disciplines are addressing fundamentally different questions (p. 1563). Knowledge regarding neurons, they write, does not shed light "on the understanding of personal meaning" (p. 1564). This does not mean that the chemical structure of paint is unimportant to a physico-chemical understanding of color or oxygenation, only that it is unimportant to the painter's *artistic* purpose in that she has no need, in order to render or create a certain kind of perceptual and affective experience, to be thinking about how molecules are linked together in specific physical patterns.

(2) In a carefully chosen clinical illustration, Yovell et al. offer a case vignette intended to demonstrate the relevance of neuroscientific information to therapeutic work.

In response, Blass and Carmeli delineate five aspects of the case description that undermine or contradict basic psychoanalytic principles (pp. 1564–1568). They conclude that the clinical intervention at the heart of Yovell et al.'s positive argument is neither neuroscientific (it is based in common psychological concepts) nor psychoanalytic (the intervention itself is not a psychoanalytic one).

(3) Yovell et al. argue that neuroscientific findings provide experimental evidence that can validate or reject analytic theories. To illustrate this point, they note that, although there is "now abundant evidence that an all-purpose appetitive drive similar to Freud's 'libido' does exist in the human brain," there is also evidence for a separate drive towards attachment that is independent of the appetitive drive (p. 1560). This, they contend, contradicts Freudian theories of development and infantile sexuality.

In response, Blass and Carmeli insist that a distinction must be made between the psychological structure of motivations and their biological substrate. Neuroscientific descriptions of brain biology are dependent, for their meaning and for their applicable relevance, on subjective experience and psychological structures, which are in turn independent of neurobiological findings (pp. 1569–1570).

(4) Finally, Yovell et al. argue that, without neuropsychoanalysis, psychoanalysis will become increasingly isolated scientifically, an assertion that is common among advocates for systematic research in psychoanalysis.

To this, Blass and Carmeli offer a clear response: If public funding and academic acceptance, and securing a fit with neuroscience, require psychoanalysis to change itself fundamentally, then public funding should be refused (p. 1570). Moreover, they suggest that the singular neuropsychoanalytic focus on *neuroscience* (rather than on history or anthropology, for example) betrays the underlying biologistic reductionism ("the special foundational status they ascribe to biology" [p. 1571]) that drives the whole enterprise. "Attributing relevance to that which is irrelevant to psychoanalysis," they write, "is harmful to the field" (p. 1571).

While these are carefully argued, consequential criticisms, I want to suggest that, by not explicitly anchoring their discussion in the central philosophical issues they have already implicitly put in play, Blass and Carmeli make it easier for the proponents of neuropsychoanalysis to keep on in their familiar position, along with its categories of language and its philosophical assumptions. Let me offer three examples.

Example 1: Yovell et al. write:

> There is now abundant evidence that an all-purpose *appetitive* drive similar to Freud's "libido" does exist in the human brain … However we also know from this same body of research that a specific drive to attachment exists in the human brain, which from the outset functions almost independently from the appetitive drive. (quoted in Blass and Carmeli 2015, 1560)

Blass and Carmeli appropriately respond by insisting on a distinction between *psychological* motivation and a *neurobiological* theory of drive. However, they do not directly confront the ontological assertion that an appetitive and an attachment drive "exist in the brain." Blass and Carmeli *imply* that "drivenness" is a term of experience and requires an experiencing subject, something missing from brain language. However, because the merging of mind and brain is ontologically fundamental in the discourse of neuropsychoanalysis, the teasing apart of these terms likewise has to take place at this fundamental ontological level. While Blass and Carmeli are getting at something crucial by merely emphasizing that psychoanalysis has different interests, by looking through a different lens (in this case, a psychological one), they do not quite untie the ontological knot at the heart of neuropsychoanalytic thinking. It may be helpful, perhaps necessary, to explicitly foreground the singular significance of the idea of an experiencing subject and of lived experience in psychoanalytic understanding, especially since this idea is critical in making a distinction between mind (which requires an experiencing subject) and brain (which is a neuronal system).[2]

Example 2: Blass and Carmeli note that neuroscientific information is irrelevant to a psychoanalytic formulation of personal meaning or to an understanding of the internal dynamics that influence the creation of personal meaning. Their focus on persons as 'meaning-creating beings' has important implications. First, it helps to define a certain notion of psychoanalytic subjectivity. Viewed psychoanalytically, human beings are embedded in an internal as well as intersubjective world of personal and shared meanings. Second, this suggests that meaning is constitutive of subjectivity, that our lived experience includes an ongoing process of interpretation and attribution of meaning. This *points* us towards epistemology, but fails to engage the pertinent ideas underlying either the neuropsychoanalytic position or Blass and Carmeli's own implicit epistemological perspective.

Example 3: Blass and Carmeli identify the biologism (often disavowed) at the heart of neuropsychoanalytic thinking. They also recognize that granting relevance to neurobiological findings when they are in fact *irrelevant* to psychoanalysis "is harmful to the field." However, why should the introduction of neuroscience into the framework of psychoanalytic thought be seen as endangering the discipline in such a fundamental way?

I would frame Blass and Carmeli's response, and elaborate upon it, in this way. Neuroscience (applied reductively) is harmful to the field because, by imposing the "brain" on the "mind," it does violence to the most basic notion of psychoanalytic subjectivity, which includes seeing subjects as situated in contexts of meaning and continually informed by unconscious fantasy and the vicissitudes of desire. By imposing a biologistic epistemology on an experientially grounded form of psychoanalytic reasoning, violence is done to fundamental psychoanalytic concepts of subjectivity and psychoanalytic knowledge, concepts that have been basic to psychoanalytic thinking and practice since Freud, and that are constitutive of psychoanalysis as a coherent discipline. These concepts are what Ricoeur calls "final referents" (Changeux and Ricoeur, 2003, 14), the fundamental concepts to which a particular discourse refers and upon which it depends in order to make sense. They cannot be swapped out or substituted for, unless one has determined that the discourse must be altered at its very foundations.

The physicalist/materialist explanation of mind or consciousness, as proposed for example by the philosopher Daniel Dennett (1991, 1996), rejects any dichotomy between subjectivist (first-person) and objectivist (third-person) description of mental

[2]M. Guy Thompson (2012/2001), in a discussion of Heidegger's phenomenology as it relates to psychoanalysis, foregrounds the role of lived experience and of an experiencing subjectivity in psychoanalytic thinking. As we will see later, Changeux and Ricoeur (2003) similarly emphasize the centrality of lived experience and of a "lived biology."

experience. In this account, consciousness (mind) emerges from physical processes in the brain, and, as a practical matter, once the materialist explanation is established, one is free to adopt an epistemological "dualism" of first- and third-person perspectives. Hence, one finds in the neuropsychoanalytic literature examples of both simple materialism and a dual-aspect binocularity that allows for an often loose juxtapositional toggling back and forth between neurobiology and languages of experience or meaning. In general, and implicitly, however, it is understood that psychoanalytic understandings of human feeling, thought, and behavior should be referenced back to neurobiology or neuroanatomy if these phenomena are to be truly comprehended, and if psychoanalysis is to survive as a respectable discipline. Blass and Carmeli assert, and to some degree explain, the irrelevance of neurobiology and its potential harm to the field of psychoanalysis. I want to elaborate on why this is so, and to do so at the level of basic philosophical questions regarding epistemology (how we come to knowledge in psychoanalysis) and ontology (the nature of mind, and of subjectivity, in psychoanalysis).

In my discussion, I will be addressing both of these elements in an effort to show how dialogue with neuroscience breaks down when actually applied to a discipline like psychoanalysis, which is so deeply embedded in subjectively lived experience and so fundamentally concerned with meaning. If one approaches psychoanalysis with physicalist assumptions about origins, by which mind and brain are merged, and then attempts to systematically interpose neurobiological description and explanation into psychoanalytic discourse, these biological reference points will become privileged modes of thinking when one encounters the inevitable ambiguities of meaning psychoanalysis both discovers and creates.[3] This is, in my view, at the heart of what Blass and Carmeli mean when they assert that neuropsychoanalysis is "harmful to the field."

The neuropsychoanalytic answer to the mind/body question

Freud: the first neuropsychoanalyst?

A conceit common to most neuropsychoanalytic authors is the idea that neuropsychoanalysis is simply a return to Freud's original intention to ground his science in neurobiology. This reversion to Freud's complicated *intentions* invariably becomes the project's passkey. The story of Freud the neuropsychoanalyst, rooted in his abandoned 1895 *Project*, has for most neuropsychoanalytic writers passed into the realm of historical fact and is frequently repeated, despite alternative interpretations (e.g. Green 2000, 81–96; Lothane 1998, 43–65; Mills 2012, 75–81). Neuropsychoanalysis will teach us, as Solms and Turnbull (2002) say, "how the mind *really* works," (p. 315) because it can now complete what Freud the neurologist could not: ground the psychical in the tissues of the brain. In this, neuropsychoanalysis presents itself as nothing less than a fulfillment, in step with continuing advances in

[3]Dennett (in Rothman 2017) responds to Chalmers' assertion of the special ontological status of consciousness, and of subjective experience, with an appeal to scientific method as the epistemological court of last resort:

> I don't see why it isn't an embarrassment to your view … that you can't name a kind of experiment that would get at "first-personal" data, or "experiences." That's all I ask – give me a single example of a scientifically respectable experiment!" (p. 55)

He espouses a kind of dual-aspect perspectivalism, but, as with neuropsychoanalysts in the fullness of their thought, his fundamental position is unimodal.

neurobiological science, of what psychoanalysis *really* is and was always *intended* to be. One can see the appeal: first, in acting as the stewards of Freud's original idea, unfulfilled due to the rudimentary neurophysiology of his day; second, in finding refuge from the bogginess of more than a century of psychoanalytic theory by turning to the promised clarities of science.

Solms (1998) summarizes the neuropsychoanalytic story of *completion* or *fulfillment* in this way:

> Freud elected to study the mind from the internal viewpoint of subjective awareness because he was forced to conclude in 1896, following the failure of his *Project*, that knowledge of its functional properties had not yet yielded to neurophysical methods of observation. However, Freud always held out the hope that *some day* the workings of the mind *would* become accessible to physical methods. (p. 8)

Moreover, Freud's epistemological commitment to scientific knowledge is evident, even in this turn toward subjectively experienced "internal" experience. After 1895, Freud

> … abandoned the false security of neuroscientific language. [He] had realized that the current state of neuroscientific knowledge was such in 1895 that his physiological and anatomical speculations were in fact *pseudo*scientific explanations, and—ironically—that he was on far more solid ground scientifically if he confined himself to a *psychological* language … . (p. 7)

What matters here for my purposes is simply that Freud's shift in 1900, with publication of *The Interpretation of Dreams*, is protected by Solms' narrative from any insinuation of an epistemological shift—although Solms does not deny that with his dream book Freud undertook a fundamentally psychological and, more importantly, *psychoanalytic* project.[4]

Omitted in this version are the epistemological implications of the embodied, subjectively experiential dimensions of Freud's new science of the mind. Solms and other neuropsychoanalysts would certainly allow that psychoanalysis is a science of the subjective and internal, but this acknowledgement of the subjective character of psychoanalytic "science" is weakened by the fact that internal subjective experience and external, material phenomena are regarded similarly.

Solms (2003) summarizes his idea in this way:

> [T]he things that we study in psychoanalysis are in fact the very same things that are studied by our colleagues in the neurological sciences. The differences in these disciplines arise only from the fact that they use different perceptual channels to study that unitary underlying thing. Psychoanalysts study the workings of the mind through internally directed perceptual channels, whereas our colleagues in the neurological sciences study it through externally directed channels. (p. 97)

Here Solms talks about the mind in a way that makes all but invisible the significance of the distinguishing characteristics of consciousness, of individual subjectivity, and of unconsciously influenced *lived* experience. Moreover, he seems to regard the "measuring instrument" in the psychoanalytic encounter to be like a microscope or other mechanical measuring device (p. 95), when it is, in fact, the totality of an experiencing subjectivity, one that is complexly, idiosyncratically organized around meanings and unconscious

[4]Kaplan-Solms and Solms (2000) also assert that Freud's turn to "psychology" was really based, not just on the limitations of the science of his time, but also on a choice he was making between then current neurological theories—from a "narrow localizationist neurology" towards a neurology based on dynamic principles, a forerunner to modern "connectionist" and "neural Darwinist" ideas (pp. 19–25).

fantasies and storylines. It is important to note that Solms' language blunts this insight by taking us away from meaning and the complexities of subjectivity, and (following certain passages in Freud) replacing these with the more rudimentary delivery system of "different sensory modalities" directed towards either the internal or external surface of consciousness. Although clearly inadequate to the richness of the psychoanalytic situation, this epistemological model nonetheless serves as philosophical bedrock for Solms' neuropsychoanalytic position.

For Solms, as for many neuropsychoanalysts, the neuropsychoanalytic narrative's "through line" is first and foremost a story of continuity, tracing the red thread of neurology that connects Freud directly to contemporary neuroscience. Thus, the invention of psychoanalysis, along with its "regrettably" isolationist development, is seen as a *detour* necessitated by an unfinished roadway, a productive digression awaiting the arrival of real science. From this perspective, in his adamant turn to the psychical and repudiation of his own 1895 *Project*, Freud was simply behaving as any responsible scientist would.[5]

Alongside the previous example from Kandel (2005), one begins to see in Solms the extent to which he assumes an epistemological commensurability or uniformity, perhaps too passively inherited from his version of the nineteenth-century neurophysiologist Freud. One also sees why this assumption has played such a foundational role in most neuropsychoanalytic writing. Solms (1998) repeatedly invokes Freud's model of scientific observation. We are equally capable of looking outward at the external world or inward at the internal, subjective world; in either case, it is consciousness being aimed in one direction or another. "In this way," he says,

> Freud embraced Kant's well-known philosophy to the effect that external reality is itself unknowable because of the distortions and omissions that flow from our perceptual apparatus, and applied it to the internal reality of the subject, which was unconscious in itself and therefore ultimately unknowable. In other words, the conventional distinction between mind and matter was shown to be spurious: *they were simply different modalities of consciousness, pointing in different directions*. And the underlying reality that these different modalities represented was *one and the same reality*. Thus, Freud was finally equipped with the *unitary* causal matrix he had been looking for, and he set about describing the internal reality of the mind as a natural sequence of events, which were no different in their essence from other natural events, and therefore subject to the same causal laws. (p. 8)

Thus, we are given an ontological sameness: mind and matter are different modalities of consciousness (dual-aspect), but one and the same reality (monism). Also, because Freud was always interested in a "unitary causal matrix" within which mental events could be viewed like natural events, psychoanalysis can be viewed as endeavoring to explore in as much depth as possible *one* aspect of this, while the view from "outside" awaited neuroscientific advance.[6]

[5]Sacks (1998) speaks for many other writers in describing the natural convergence between psychoanalysis and neuroscience. Commenting on Edelman's neural Darwinist ideas, Sacks says, "The attunement here between Freud's views and Edelman's is striking—and here, at least, one has the sense that psychoanalysis and neurobiology can be fully at home with one another, congruent and mutually supportive" (p. 21).

[6]Kant's idea of intrinsic or transcendental forms (of knowing and experiencing) becomes the basis for many post-Kantian developments in Continental philosophy, from Hegel to Foucault, but including Husserl, Heidegger, and hermeneutic phenomenology. In his use of Kant, Solms elides one of the primary insights of this philosophical tradition: that the pre-given categories of subjective experience should be foregrounded, should lead us not merely to the facticity of the object, but to the forms of experience by which that object is known. From this perspective, the phenomenology of lived experience is first of all the basis for, not the object of, empirical or scientific observation.

The argument works as an historical platform for Solms' assertion of a "dual-aspect monism" with respect to the mind/body question. It is also more palatable than a naïve physicalist position, on the one hand, or Descartes' substance or property dualism, on the other.[7] In the end, in Solms' neuropsychoanalytic story, not only is Freud the first neuropsychoanalyst, he is also the first neuropsychoanalytic dual-aspect monist, like Solms himself. Mind and brain are the same stuff, viewed from different perspectives—and at least in this version, with its depiction of Freud's aspirational allegiance to the rules and methods of traditional science, any epistemological or philosophical contradiction is blurred from view.[8]

Dual-aspect elephants

The equivalence neuropsychoanalysis makes between the mental and the physical (they are aspects of the same underlying reality) follows from the epistemological position that each is being studied by the same observing consciousness through different perceptual channels. As disciplines, the science of subjectivity and the science of objective neurobiological structures, the effort to look from inside or from outside, become "aspects" or modalities of a common endeavor. However, Solms goes a step further. When mind and brain are seen as "no different in their essence … and therefore subject to the same causal laws," then what is different about the *particular character* of the object of study (in this case the mental and particularly unconsciously determined experience of an intentional subject) is made both ontologically and epistemologically inconsequential. The Kantian framework Solms invokes endorses similarity: there is nothing special in Freud's assertion that the internal unconscious realm is unknowable, since the external world is equally unknowable. Thus, equivalence is asserted in both registers—how things *are*, and how things are *known*.

As Karlsson (2010) remarks:

> The basic problem is that the subjective side and the bodily side (the brain) are two radically different perspectives that cannot be connected or bridged. … This circumstance—that subjective consciousness is a first-person inside perspective, while observation of the brain is a third-person external perspective—is recognized by Solms and Turnbull, *but this recognition does not seem to entail any obligation*. They do not take this unbridgeable gap between these two perspectives seriously. (p. 52, italics added)

This lack of *epistemological obligation* is sometimes justified by the allegory of the blind men and the elephant, which is invoked by neuropsychoanalytic writers to illustrate the need to have multiple perspectives from which to describe the mind/body elephant, since one perspective can never grasp the great animal's full nature. The allegory is flawed in two simple ways. First, the blind investigators, whether touching the tail or the trunk, are gathering information with their hands—that is, in the same way, by touch, although perhaps from different angles. Second, one assumes they must in the end refer back to an overarching or common epistemological framework by which all information they have gathered can be made sense of. However, this crucial step is not described. The pluralism of first- and third-person perspectives is *suggested*, but synthesis

[7]For brief descriptions of different positions relative to the mind/brain question, see Nagel (1995b) and Chessick (2001).

[8]Although rhetorically compelling, the appropriation of Freud's presumed intentions for the future direction of his psychoanalysis constitutes an obvious fallacy. Psychoanalysis was never confined to Freud's (often conflicting) intentions, nor was it frozen in place in 1939. Freud could not foresee the philosophical conundrums yet to be presented to future readers by his invention. This has been left for us to sort out.

is never really *explained*, and this allows an underlying sameness to be assumed. Epistemologically it is really just *one* person assessing all the parts of *one* ontologically constant elephant and then combining what he has gathered in one way. Hence, the true epistemic dualism or pluralism intrinsic to an appropriate allegory is avoided (cf. Karlsson 2010, 52–53).

Dual-aspect monism purports two sides, but, no matter how fully developed, a first-person perspective cannot be *synthesized* with a third-person perspective. Toggling from one to the other does not count as synthesis. I would contend that one cannot simultaneously listen psychoanalytically and neurobiologically. Correspondingly, a patient cannot be moved by a description of their neurobiology unless and until that description has meaning to them, at which point it has become contextualized as an aspect of their personal, subjective experience and is no longer being talked about in a neurobiological way. It has been appropriated, re-contextualized, and the neurobiological referent has been displaced.

There is another important point to be made here. As I've suggested, the rules governing any dialogue between neuroscience and psychoanalysis must be guided by some epistemological framework or other. Also, if the guiding perspective is scientific it can only result in a creeping imposition of that framework's epistemological assumptions on psychoanalytic notions of subjectivity, thus contriving to "reduce" objectively what is ontologically irreducible (subjectively lived experience).[9] Any dialogue between psychoanalysis and neuroscience *itself* requires a reasoned process of assessment, appreciation of constraints, and acknowledgement of fundamental differences (a hermeneutic), as well as an appreciation of the epistemological obligations required in making psychoanalytic subjectivity one's subject matter. It is truer to the realities of how we work and think psychoanalytically to say that neuropsychoanalysis is not looking at the same elephant at all, but rather at two different elephants—a first-person elephant and a third-person elephant—and wondering how it can merge them into one elephant. Karlsson seems to suggest this can't be done as proposed without making a bloody mess of at least one of them,[10] and I am inclined to agree.

Nagel (1986) remarks that to attempt to explain the mental as one would non-mental phenomena is "both intellectually backward and scientifically suicidal" (p. 52). Arguments from authorial intention and those employing epistemological elephants, by obscuring important differences, are often being used towards precisely this end. In order to create the grand reductions implied by such terms as a "neurobiology of meaning" (Modell 2003) or a neural "ethics" and "physics of introspection" (Changeux and Ricoeur 2003, 68.), neuropsychoanalysts simultaneously *acknowledge* and *overlook* fundamental

[9]Science is of course interpretive; humanistic disciplines do indeed rely on empirically derived knowledge (in the broad sense of "empirical"); and psychoanalysis beginning with Freud has trafficked in both realms. However, calling psychoanalysis an epistemologically "hybrid" discipline does not, in my view, excuse it from the need for an overarching (dialogical, practical, reasoned) framework if it is to be coherent.

[10]This is not an argument against interdisciplinary dialogue, only an insistence that differences in basic referents are acknowledged at the start and preserved as fundamental to all further discussion. See Ricoeur (Changeux and Ricoeur, 2003, 121). One could object to my emphasis on philosophical hermeneutics and phenomenology when Pragmatism, Wittgenstein, Davidson, or even phenomenologically oriented, empirically grounded consciousness studies (Chalmers 1996, 2004, 2010; Varela and Shear 1999) provide alternative philosophical approaches. I would suggest that, whatever approach one takes, one has to reckon with how scientific knowledge and process are to be considered and then brought into the realm of subjectively constructed meaning and intention. It is hard to see how this relationship can ever be self-evident or epistemologically uncomplicated.

obstacles to neuroscientific aspirations.[11] Most fundamentally, such terms fail to appreciate what Ricoeur describes as our "lived biology" (Changeux and Ricoeur, 2003, 74) and the experiential allusiveness of the psychoanalytic unconscious.[12]

While the twentieth century saw a variety of strategies for putting the mental into the physical realm and undoing the dualism initiated by Descartes and Galileo (Nagel 2012), a certain "dualistic" sensibility persists in all of us. Lived experience requires corporeality, yet at the same time our lived experience cannot be reduced to (or fully captured by) the body or the language of biology. Our sense of having or being a separate mind[13] needn't be seen as residue from Descartes' substance dualism (matter, spirit) or radical skepticism about the existence of other minds. It can be seen instead as evidence of the powerful claim made on us by our *immediate* experience of ourselves in the world—that is, of one's otherness, of oneself over against the presence or absence of everything else. The strictly neurobiological perspective would say that the mind/body dualism persists due to a lack of knowledge of the biological underpinnings, and that a monadic solution, a grand integrative recoupling of mind and brain, awaits the evolution of neurobiological science.[14] However, life-as-experienced always resists absorption into a unifying objectivism, whether empirical, rationalist, or biological. The phenomenology of our lived experience, especially when conceptualized psychoanalytically, is more than you or I can know or explain; it always harbors a latent, irreducible otherness and excess, a remainder of Being.[15]

Neuroscience and the experiencing subject

Freud focused on instincts being the primary propulsive forces driving the development of desire in relation to its manifold objects. The drives were to be understood as psychical forces at the interface between mind and body; they were the representatives of our embodied natures. Moreover, Freud's *psychoanalytic* body, properly understood, was not a collection of physical organs, but the body as subjectively experienced, and in this it was the very seat of our humanness. Loewald (1971) gives us an almost Whitmanesque description of the psychoanalytic body: "The life of the body, of bodily needs and

[11]Merleau-Ponty (1960) asks a related question: "Does not the notion of libido lose all its meaning if there is a 'cellular libido,' if it is a property of the cells … ?" (p. 68).

[12]I am not making an argument for or against a traditional Freudian notion of the unconscious. A more intersubjectively constructed unconscious as proposed by Binswanger (1963), for example, which still includes an *Eigenwelt* or "private world of self" (see Frie 2004, 35) while replacing repression with "unformulated experience," can also serve to undergird the centrality of lived experience and its elusive and allusive character.

[13]Hass (2008) suggests something like this in his study of Merleau-Ponty: "There is … a *distance* between the experiencing self and behaviour which is not a transcendental chasm or a dualism, but precisely a separation (*écart*) … This separation, this dehiscence, is why I am never *quite* at one with myself. It is why, despite all our corporeal and conversational reciprocity, there is irremediable distance between oneself and others" (p. 111). This notion of *écart* expresses the experience of *difference* without dichotomy or dualism.

[14]Opatow (1999) articulates his version of such a neuropsychoanalytic solution. He writes, "Only very recently has the mind-body problem fallen from the precincts of philosophy and theology into the purview of standard empirical science. The scientific project is grounded axiomatically in a unitary ontology of mind and matter, and in the mandate that this unity be explicated progressively … [O]ur present methodology employs a provisional conceptual apparatus that, admittedly deficient (i.e. dualistic), will be used as the instrument for its own superseding" (p. 97). As he says, "Settling this issue would seem a straightforward matter of diligent empirical research" (p. 108).

[15]I expect that Hanly (1999, 2006, 2014a, 2014b) and Hanly and Fitzpatrick Hanly (2001) would object that I am perpetuating a confusion between *psychological* subjectivism and *epistemological* subjectivism. However, what Hanly sees as confusion might be seen instead as an assertion of the inextricable relationship between the forms of subjective experience and the character of what and how we know things.

habits and functions, kisses and excrements and intercourse, tastes and smells and sights, body noises and sensations, caresses and punishments … ," and concludes, "all this is the body in the context of human life. The body is not primarily the organism with its organs and physiological functions, anatomical structures, nerve pathways, and chemical processes" (p. 125). In a way that evokes Merleau-Ponty's (1945, 1964) phenomenological notion of experienced corporeality, Loewald (1971) is telling us that it is *this* body, the *lived* and *fantasied* body, that is the fundament of human experience as conceived psychoanalytically, and of psychoanalytic theory.[16]

Given this argument, we are obliged to ask what a neuropsychoanalytic notion of subjectivity might be. André Green (2002) observes, "More often than not, this definition [of the subject in neuropsychoanalysis] is neglected or hidden" (p. 69). Karlsson (2010) goes a bit further in suggesting that, in neuropsychoanalytic writing, the subject as subject is left out altogether: there is no meaningful interest in the experiencing subject, except as a resource for terms that can be imported into an otherwise subject-less brain language.

Returning to Ricouer's point, then: To say that "the brain thinks" is to excise the experiencing, thinking subject, while simultaneously proceeding as if the term "brain" included an inhabiting subjectivity, one that "thinks." Bennett and Hacker (2007) make a complementary point: the neuroscientific attribution of psychological qualities to the brain makes no sense. "[T]he brain," they write, "cannot be conscious, only the living creature whose brain it is can be conscious—or unconscious. *The brain is not a logically appropriate subject for psychological predicates*. Only a human being and what *behaves* like one can intelligibly and literally be said to see or be blind, hear or be deaf, ask questions or refrain from asking." (p. 21)[17]

Perhaps one might say, along with Solms, that psychoanalysis and neuroscience are looking at the same thing differently. As Solms (2003) asserts, "the feeling of depression and the neurochemical process that corresponds to it are in fact two different perceptual representations of *the same underlying thing*" (p. 97). However, can they really be versions of the same thing when a neuronal model cannot accommodate an experiencing subjectivity? This is, after all, hardly a trivial difference. It seems more accurate to say that the "subject matter" for each is different by virtue of being constructed from, as Loewald (1971) says, such "different modes of experiencing and ordering reality" (p. 134).

Embodiment is not mere corporeality; it is a fundamental category or condition of experience. Moreover, our embodiment occurs within pre-existing contexts of cultural and linguistic forms. These experiential conditions of ordinary life are, from a phenomenological/hermeneutic perspective, primary. They constitute a foundational, first-order dimension of human knowledge, not one that is secondary to the findings of neuroscience. In fact, they afford neuroscience a context, a world within which to make sense. Only when we properly understand what a neurobiological explanation can and cannot explain, and when the differences between realms is clear, can the findings of neuroscience be evaluated as potentially relevant or not to psychoanalysis. However,

[16]Merleau-Ponty (1960) comments: "What we learned from all the material drawn from dreams … was to discern an imaginary phallus, a symbolic phallus, oneiric and poetic" (p. 69).

[17]Bennett and Hacker (2003, 2007) make several detailed philosophical arguments (some of which are complementary to Ricoeur's) against the pervasive, illegitimate neuroscientific attribution of psychological and experiential properties to the brain or hemispheres of the brain that should only be attributed to "the whole animal" or experiencing subject.

this is only possible under the bright rubric of epistemological difference and discursive independence. A cat is not a battleship. A mind is not a brain.

False dialogue: the confusion between epistemology and methodology in neuropsychoanalysis

On the issue of merging psychoanalysis with neuroscience, Kandel (2005) observes that "classical genetics and molecular biology have merged into a common discipline, molecular genetics. … So be it with psychoanalysis" (p. 94). In other words, they are all epistemologically the same, even if they may be methodologically different (see also Changeux and Ricoeur, 2003, 242). Such a view seems to follow naturally from a dual-aspect perspective on the mind–body issue, as we have seen in Solms' use of Kant. Once it has been established that psychoanalysis ought to be concerned, as Solms' Freud was, with a unitary mind/body matrix, the convergence between psychoanalysis as the science of subjective mental experience, and neuroscience as the science of the brain, is all but a *fait accompli*.

The necessity of this conclusion is expressed in different ways. Based on the assumption that exploration of the subjective character of mental experience is insufficient, it follows that psychoanalysis requires neurobiological and experimental findings in order to establish relevance, validity, and clinical value. Hence, it is further assumed that, both theoretically and clinically, psychoanalysis has gone as far as it can on its own. As Solms (2000) writes, psychoanalysts who fail to assimilate advances in neuroscience "will be increasingly marginalised both scientifically and professionally, and will be unable to participate in this important intellectual revolution" (p. iii). The unspoken overarching assumption in all of this is that psychoanalysis would do well to reconstruct itself as a comprehensive, clinically and theoretically pluralistic discipline, a convergence of biological and psychodynamic psychiatry, cognitive neuroscience, and academic psychology. In this view, it is insupportable for psychoanalysis to be merely psychoanalytic, with its own conceptual architecture and rules of practice, and for psychoanalysts simply to practice according to this framework. It has even been suggested that such provincialism is tantamount to malpractice (Westen, quoted in Safran 2012, 717).

Fotopoulou (2012b) describes the necessary rapprochement in this way:

> [N]euroscience cannot and should not directly determine how psychoanalysts and psychodynamic psychotherapists treat their patients at the level of "the personal." However, in time, neuroscience can influence the universal metapsychological models that are put forward, discussed and debated within psychoanalysis, and then applied, and in this sense neuroscience can *indirectly* influence the fate of psychodynamic therapies. … Given that neither of the two facets [neuroscience/psychoanalysis] is sufficient to fully describe the actual phenomenon (the so-called "mindbrain" entity), collaboration and dialogue may *constrain and enhance each other's models*, without incorporating or eliminating each other's unique scope and practice. (pp. 22–23)

In neuropsychoanalytic writing, "dialogue" or interface is often seen in just this salutary and unproblematic way, and on its face such a view might seem perfectly reasonable. However, Fotopoulou's language is at odds with itself. How can psychoanalysis be "constrained" by the findings of neuroscience without elements of psychoanalytic thought, and the elements of practice grounded in those ideas, being "eliminated"? Isn't that the whole point of "constraint"? And how can psychoanalysis be "enhanced" by the findings of neuroscience without it "incorporating" aspects of neuroscientific theory? Isn't that also part of the project's whole purpose?

Such a stance asks that we accept propositions that oppose one another. First, we are given the idea that psychoanalysis would and should be changed by convergence with neuroscience, but at the same time we are told that such convergence will not change psychoanalysis fundamentally. A *radically* different psychoanalysis will emerge, according to Solms, but it will not be *fundamentally* different, according to Fotopoulou. How this is possible is not spelled out.[18] Second, the model of dialogue suggests an acknowledgement of epistemological differences governing the different disciplines, which contributes to their uniqueness, but we are asked to accept on faith that this divergence poses no complication, and that no overarching *epistemologically coherent* framework needs to be articulated in order for such dialogue to be potentially productive. Knowledge is treated generically, whether it is knowledge of the tail or of the trunk, so there is no impediment to neuroscientific and psychoanalytic knowledge talking together, constraining and enhancing, eliminating and incorporating.[19]

Fotopoulou proposes a dialogical framework, but it is only vaguely drawn and mixes wildly divergent ideas. By way of example, she (Fotopoulou 2012a) invokes Gadamer's hermeneutic notion of "horizons of understanding" (the situated and contextual nature of all understanding), acknowledges the "epistemological gulf" that exists between psychoanalysis and neuroscience, but then calls for "establishment of a clear and adequate conceptual framework that can be shared across different disciplines. Once such a framework is in place, alternative explanations can be translated into a common language and can be evaluated on empirical grounds" (p. 5). This is like Opatow's (1999) aspirational, empirically grounded bridging discourse, of course. One feels the slippage, unexplained, from the notion of dialogue without any need for agreement, into "translation," "a common language," and "empirical evaluation."

It certainly seems as if epistemologically divergent knowledge is being collapsed into an empirically validated, unimodal kind of knowledge, despite the fact that Fotopoulou's framing description insinuates something quite different. She seems to endorse the dialogical character of an anti-reductionist hermeneutic (Gadamer 1960, 1981), and proceeds under the pretense that dialogue needn't risk alterations of epistemic commitments. However, if difference is maintained, can real dialogue of this sort actually *avoid* epistemological violence of some kind? Can the objectivism of neuroscience "dialogue" with the subjectively grounded interpretive epistemics of psychoanalysis without it resulting in either incoherence or important epistemological accommodation? Because the knowledge she is talking about is not all the same; the psychoanalytic mind, as I've argued, is a very particular elephant.

In the end, neuropsychoanalytic integrationists do not regard the difference between mind (which is grounded in subjective experience) and brain (which is an objectively described neurobiological entity) to include difference at the level of fundamental ontology. However, this disregards the idea that, because each exists as a different category of being, a

[18]Pulver (2003) also argues that partnership with neuroscience will not affect psychoanalytic *technique* or how we talk *personally* with patients.

[19]Mayes' (2005) neuropsychoanalytic description of the dialogue between the "two epistemologies" of developmental science and psychoanalysis is another influential example of this problem, where a methodological difference is misrepresented as a meaningful epistemological distinction.

different kind of entity, each can be known and talked about only in certain ways, which means they *cannot* be known or talked about in other ways. To this point, Mills (2012) remarks specifically on the "mereological fallacy," also discussed in Bennett and Hacker (2003, 2007), operating in the neuroscientific position. He describes this as "a fundamental attribution error where one ascribes the acts or characteristics of a whole to its parts," thus reducing the human subject to "material-efficient reductive forces" (fn, p. 53) or, in the case of neuroscience as applied to psychoanalysis, reducing the mind to the brain.

There is indeed a material substructure and physical laws underlying every moment of psychoanalytic experience. Also, there are physical patterns in the brain that correlate (however complexly) with these "internal" subjective experiences. However, our experience is not of neurobiological patterns; it is rather of the mental events that accompany these patterns, the feelings and thoughts, the rhythms and meanings.

When a feeling of existential dread washes over a patient, they might feel for a moment that they can't breathe, but then they breathe, and then they talk, and perhaps the dread subsides. However, without a language of dread, the language of receptors has no referent, tells no story worth hearing. "It is worth remembering," Dominique Scarfone (2012) remarks, "that there are no 'words' in Broca's area and no 'fear' in the amygdala, although these brain structures are vital for speech and for feeling fear, respectively" (p. 1288). Languages represent different ways of ordering reality, and sometimes different ways of determining what "reality" is exactly, and then of coming to knowledge about that reality. While neuroscience is the realm within which research into neurological disorders is possible, it is generally not helpful in our talking about the *meaning* of anything, not even the meaning of neuroscience (whether in a personal, cultural, historical, or psychological sense), without resorting to a separate discourse designed to capture cultural or personal meanings.

The discourses of mind and brain: a *semantic* dualism

The languages of mental and physical experience offer very different ways of looking at oneself and the world. In saying this, we make an important shift. Although the brain is a "thing," a definable physical system, the only way we can know about it and talk about it is by way of a kind of discourse, a brain discourse. So we have mind discourses, and we have brain discourses (Brothers 2002), and we use these different languages to talk about perception, thought, memory, affect, or about our sense of self, meaning, and value, all the ways in which we experience ourselves in the world. Ricoeur (Changeux and Ricoeur 2003) expresses the difference very clearly:

> These discourses represent heterogeneous perspectives, which is to say that they cannot be reduced to each other or derived from each other. In the one case it is a question of neurons and their connection in a system; in the other, one speaks of knowledge, action, feeling—acts or states characterized by intentions, motivations, and values. I shall, therefore, combat the sort of semantic amalgamation that one finds summarized in the oxymoronic formula, "The brain thinks." (p. 14)

Yet such amalgamation is endemic not just to neuropsychoanalytic writing, but to contemporary psychoanalytic discussion more broadly, a consequence of the widespread assumption that brain language is applicable to any realm of psychoanalytic knowledge —it should be permitted, that is, to go wherever it pleases. When Damasio (2010) suggests that the principal objection to assuming equivalences between brain processes and mental events is that "to date science has not been able to determine the physical attributes of mental states" (pp. 335–336), he intends that the solution is at hand, which assumes the reducibility of subjective mental experience to brain language or some hybrid discourse. In this, I would suggest (adopting a phrase from Nagel's [1995a] critique of Dennett), he "misses the real problem from the start" (p. 86).

As Blass and Carmeli (2015, 2007) suggest, seeing mind and brain discourses not just as different ways of looking, but as concerned with different objects (or subjects) of study, appropriately complicates the causal linkages between them. Just as there is no apparent solution to the problem of consciousness (what is it, and how do we describe not just its nature, but the simple fact of its existence?), there is no reductionist answer to the irreducible "ontological density" (Ricoeur 1973, 75) of mental experience. As Nagel (1986) says, "The subjectivity of consciousness is an irreducible feature of reality—without which we couldn't do physics or anything else—and it must occupy as fundamental a place in any credible world view as matter, energy, space, time, and numbers" (pp. 7–8). Neurobiology cannot answer the *philosophical* problem posed by consciousness, nor can it solve the philosophical riddle posed by the mind/body relation. In the end, neither question is a neurobiological question.

I do not oppose objective methodology in the service of explanation, but specific important qualifiers should be considered. First, it is understood that systematic research, most especially in the social or human sciences, emerges from prior contexts of meaning, some entirely hidden or outside of awareness, some simply by necessity excluded for the purposes of thinking in a certain way. Second, in coming to interpret and understand the findings of systematic research, any number of possible perspectives might be brought to bear, and the findings will be interpreted in any number of different ways. This is a process of critical reasoning, not of objective, empirical study.[20] So, prior to systematic research, in its organization and delimitation, and at its conclusion, in the variegated interpretations of its findings, objective study is framed by interpretive activity. And *psychoanalytic* interpretive activity, psychoanalytic *thinking*, is especially rooted in individual, affectively colored, unconsciously influenced subjective experience. This is where psychoanalytic thinking begins and where psychoanalytic thinking ends, wherever else it might travel in between.

Ms. B begins her session elaborating on her struggle to break free from entanglements with her mother and siblings, an incremental and painful process made worse by the incestuous and sadomasochistic bonds that held the family together and insinuated themselves into Ms. B's sense of self. She spontaneously shares a memory. She is quite young, perhaps five or six or seven, and it is winter. She has decided, for reasons she can't recall, to venture outside, and (in her recollection at least) she dresses herself in snowsuit and boots, gloves and hat, determined to be outside in the white expanse of yards and houses. The snow is very fine, swirling fiercely, biting her cheeks, with drifts past her knees. She manages to slog to her neighbor's back yard. She looks up only to see white and gray, the houses down the street vanishing into shadows behind the blur of snow. She looks inside a snow fort but it is empty except for a single half-buried rubber boot. She remembers her feeling of utter

[20]Sometimes advocates for objectivism in a normal psychoanalytic process want to refer to such clinical reasoning as part of a *scientific* activity, thus eliding any difference between a complicated *lived* engagement which includes practical reasoning, and a scientific process of hypothesis formulation and testing.

aloneness, how desolate this familiar landscape suddenly seemed to her, not a single other moving human form anywhere. Shivering and exhausted, she surrenders and goes back inside. She feels damp and unhappy, and her toes are numb, and this scares her. When her mother finds her on the floor of the mudroom she says, "What is wrong with you!" and sends her to her room. Now everything is damp, her hair, her underwear, so she takes everything off and crawls into bed with a towel wrapped carefully around her head so as not to get the pillowcase wet. She thinks she will never feel her toes again, and she decides right then that she will never tell anyone about it, not even the doctor.

As any reader of Proust or student of contemporary neurocognitive science knows, memories are reconstructions shaped by the moment in which they are made or remade. Our memories are notoriously unreliable narrators. According to Gadamer's (1960) notion that experienced meaning can only be prejudicial, can only occur within particular contexts of meaning and value, our minds and memories are *necessarily* unreliable in the sense of incomplete, framed, idiosyncratically focused, pre-organized. Without the prejudices afforded by a particular horizon of experience, a point of view, we could not make sense of anything, and knowledge of the sort that interests us psychoanalytically would not be possible. Hence, learning about the neural correlates of our identifying projections, our capacity to feel the light or the wind, to experience desire or aloneness, adds nothing of *psychoanalytic* interest in deepening our understanding of the feelings and meanings of Ms. B's memory. Seeing which regions of her brain light up on a PET scan when she experiences an ache of beauty is, in the context of our clinical encounter, of limited interest to her. This brings to mind Ricoeur's (Changeux and Ricoeur 2003) remark: "when I am told that I have a brain, no actual experience corresponds to this; I learn about it in books" (p. 16).

Assuming the absence of significant organic pathology, the recollections, silences, defensive shifts, changes in mood or sensation, daily recountings of ordinary events, formulations of meaning or value, graspings after what exceeds articulation, are listened to in a psychoanalytic context much as one might hear this described memory. We are each drawn idiosyncratically into our own or our patients' subjectively colored recollections, and yet the meanings and evocations for each listener will change and rearrange as the moment of vantage point shifts, much as the speaker would tell the memory differently depending on where she is situated, or would feel moved just once or rarely to think it at all.

As Karlsson (2010) points out, "the subject, in terms of a meaning-bestowing, intentional subject, disappears. … in neuropsychoanalysis. And, in consequence, the *world* is also missing, with all that goes with it—other human beings, history, future projects, and a relation to oneself" (pp. 55–56). This is not intended critically, except to the extent that neuropsychoanalytic writers are inattentive to what is missing and why, and to the complications of bringing neurobiology into relation with the "world" as understood psychoanalytically.

The neuropsychoanalyst will often use the word "world" and assume they are engaged in dialogue using a shared language. Fotopoulou (2012a) does this when she invokes Gadamer's idea of "horizons" without at all meaning what Gadamer means. Ricoeur makes this point to Changeux in this way:

> You employ the term *world* too freely. The world is not only the immediate environment; it is the horizon of a total experience. This notion of horizon is perhaps exactly what is eliminated in the construction of the mental object in order for this object to furnish an equivalent to what you have constructed in the neuronal field (Changeux and Ricoeur 2003, 92).

Ricoeur remarks on the fact that neuroscience relies illegitimately on "metaphysical anthropomorphisms" (Changeux and Ricoeur, 2003. p. 89) and semantic amalgamations as bridging terms. The neuroscientist wants to traverse from a neuronal assembly to an attribution of meaning, and create the sense that the one follows from the other, by mixing languages without acknowledgement of the fundamental distinction between objective and subjective experience. Ricoeur says bluntly that, if the term "mental object" is intended to link the physical and the mental, "Then it's an illegitimate term" (p.101). It is illegitimate because it presumes to synthesize what are semantically divergent discourses and to circumvent the gap, the *semantic* dualism separating brain and mind discourses, merely by constructing a hybrid term. Of course, "neuropsychoanalysis" might serve as another example.

I understand this to mean that, while it is true that the "dual-aspect" of a neuropsychoanalytic epistemology should not be misconstrued as an *ontological* dualism (we can of course look at the same thing from different perspectives), neither should a monistic ontology (the proposition that mind and brain are a single entity) be used to legitimize the merging of different "aspects" or *epistemological* systems. Ricoeur encapsulates his approach in this way:

> I proceed, then, from a semantic dualism that expresses a duality of perspectives. … Mental experience implies the corporeal, but in a sense that is irreducible to the objective bodies studied by the natural sciences. Semantically opposed to the body-as-object of these sciences is the experienced body, one's own body. … I do not see a way of passing from one order of discourse to the other: either I speak of neurons and so forth, in which case I find myself in a certain language, or I speak of thoughts, actions, and feelings that I connect with my body, to which I stand in a relation of possession, of belonging. (pp. 14–15)

Ricoeur's point is simple, but full of implications. Ordinary experience, protean and dynamic, always outstrips any scientific effort to capture it; and the phenomenology of subjective experience, once understood, carries with it obligations that must be honored in the formulation of psychoanalytic ideas about subjectivity and psychoanalytic knowledge. Only when the distinction between basic referents is maintained, and the pull toward sameness (in the registers of knowledge and of Being) is unyieldingly refused, should psychoanalysis undertake an assessment of the relative value of neuroscientific information.

Somewhat unexpectedly, perhaps, philosophical hermeneutics and phenomenology return us to the Freud of the *Project*, but from a perspective quite opposite that of Solms and neuropsychoanalysis. From this vantage point, Freud's neurobiological speculations, to the extent that these created initial maps for his theories of the mind, turn out to be proto-psychoanalytic narratives whose true value rests in their (broadly speaking) metaphoric and conceptual effort to capture and understand the mental as corporeally lived experience, as embodied subjectivity in the context of a lifeworld. The emphasis here is not on drive as a mere expression of biology, but on drive as a *mental* event grounded in biology. The neurobiological particulars of the organic substrate are, in my view, relevant primarily as guardrails that help direct a larger process of psychoanalytic exploration; what matters psychoanalytically are simply the various ways in which we experience ourselves as embodied, desiring subjects, not the neurobiology of that experience.[21]

Conclusion

The preferred position among neuropsychoanalysts is some form of epistemological pluralism or dual-aspect monism, although the shadow of an unavowed biological reductionism often lurks beneath the surface. One sees evidence of this in the slippage common in neuropsychoanalytic writing, where epistemological differences are divested of epistemological obligation. As I have tried to show, this has potentially devitalizing implications for how psychoanalysis regards itself as a discourse, how it understands the nature of psychoanalytic knowledge, and how it conceives of the psychoanalytic person or subject.

In my view, this position carries us to the threshold of granting privilege to scientific or objectivist explanation. This is readily evident in neuropsychoanalysis' cardinal assumptions: the sense of urgency in needing to correct psychoanalysis' isolation from mainstream science; an epistemological bias towards explanations grounded in biology; and a conviction that neurobiological science, before any other discipline, should be employed to rescue psychoanalysis from the body of flabby guesswork known as psychoanalytic theory. By acceding to these assumptions, psychoanalysis opens itself to unwarranted, yet radical, revision, based on neuroscientific theories and findings.[22]

My response to urgent calls to interdisciplinary pluralism and charges of negligence is simple. Psychoanalysis is not a comprehensive mental health treatment or general psychology. It is not a branch of psychiatry or biology. It is not required, by reasons of science or practical economics or cultural preference, to reconstruct itself according to the rules of other disciplines organized according to theoretical, methodological, or epistemological principles and assumptions different from those of psychoanalysis. While not opposed to interface or dialogue with related disciplines, psychoanalysis should insist on privilege within its domain, should be "epistemologically careful" (Scarfone 2012) and unapologetically protective of its right to interpret critically whatever comes to light in its interactions with neuroscience, to decide what matters and what has no interest.[23] The relationship of one to the other is a discussion, offering a spectrum of opinion, and certainly not *necessary* to the practice of psychoanalysis.

[21] I borrow the notion of a "guardrail" from Westphal (2011) in his discussion of Ricoeur's use of objectification.

[22] Thompson (2012/2001) reminds us of Heidegger's conviction that modern culture is dominated by an "ontic," objectifying form of thinking which often only alienates us further from the reality we intend to know—in the case of psychoanalysis, from the unconscious beliefs and fantasies undergirding and informing our experience. Heidegger asserts an "ontological difference" between the "ontological" (the fundamental forms by which the world is disclosed to us) and the "ontic" (the empirical "objects" themselves which populate the world). Psychoanalysis is fundamentally concerned with the ontological aspects of human being—temporality, care, the ongoing tensions between loss and possibility, anxiety and the avoidance of suffering, being-towards-death and its avoidance—as constitutive of our humanness. Science is concerned with the ontic, with worlds of objectively observable things. As Sass (2015) remarks, one consequence of forgetting the ontological difference … is the tendency to conceptualize even consciousness or subjectivity itself as what Husserl called a "tag end" of the world—namely, as something that, whatever its specialness, nevertheless has the status of another entity *within* the world rather than as the condition for the world, as the grounding for "worldhood" itself. (p. 424)

[23] By this I mean psychoanalysis as a *discourse*. As Brothers (2002) points out, analysts in practice might function as psychiatrists, or might refer an analytic patient to a psychiatrist, neurologist, or other specialist. However, when functioning as *analysts*, they are operating in a specific realm and in a particular way.

Psychoanalysis tells a *psychoanalytic* story about the progress of desire through the filaments of ordinary human experience and in a language of its own, organized by theoretical concepts that are to a significant extent uniquely psychoanalytic. For Ricoeur (1970) psychoanalysis includes a "semantics of desire," which brings with it a notion of a complexly constructed desiring subject with a "vocation" for meaning and for language.[24] This is at the heart of psychoanalysis, and what neuroscience has to say about it is often going to be either inherently reductive (the terms of psychoanalysis being reduced to, or re-transcribed to fit with, the terms of neuroscience), or simply beside the point. This is true in part because, from within the frame of neurobiology, one has only a mediated, indirect recourse to psychoanalytic ideas and language. Although astonishing, even revolutionary, neuroscience is something else altogether, a different epistemological animal, one peculiarly unequipped to grasp the "ontological density"—the meanings, affective coloring's, and unconscious fantasy constructions—of our subjectively experienced "lived biology." Neuroscience cannot go where psychoanalysis lives, cannot speak to its most salient parts, and this fact alone argues for the necessity of a respectful, sensibly cautious relationship. Any interrelation between neuroscience and psychoanalysis warrants care and suspicion. There are far-reaching, potentially grave dangers for psychoanalysis when neuroscience is misrecognized in its fundamental differences, and is injudiciously employed to answer questions that properly belong to the language and conceptual architecture of psychoanalysis.

Translations of summary

En tant qu'entité ostensiblement hybride, la neuropsychanalyse nous invite à jeter un regard critique sur ses hypothèses relatives au savoir et à la subjectivité afin de parvenir à comprendre la relation qu'entretiennent ses moitiés sans trait d'union. L'auteur de ce article examine les différences entre l'esprit (qui est fondé sur l'expérience subjective) et le cerveau (qui est une entité neurobiologique décrite objectivement) et considère que les neuropsychanalystes ont tendance à la fois à reconnaître et, dans un second temps, à négliger la nature unique et irréductible de l'expérience vécue, ainsi que les différences fondamentales entre l'esprit au sens psychanalytique (qui nécessite un sujet de l'expérience) et le cerveau (qui est un agrégat de neurones). S'appuyant sur une base philosophique, l'auteur soutient que la psychanalyse se trouverait potentiellement menacée par les neurosciences si l'on méconnaît les différences fondamentales qui les caractérisent et qu'on les utilise sans discernement en partenariat avec la psychanalyse afin de répondre à des questions qui appartiennent en propre au langage et à l'architecture conceptuelle de la psychanalyse.

Als unverkennbar hybride Entität fordert uns der Begriff „Neuropsychoanalyse" dazu heraus, die ihm inhärenten Annahmen über Wissen und Subjektivität kritisch zu untersuchen, wenn wir verstehen wollen, welche Beziehung zwischen seinen ohne Bindestrich zusammengefügten Hälften besteht. Der Autor untersucht die Unterschiede zwischen Geist/Psyche (in subjektiver Erfahrung gründend) und Gehirn (einer objektiv beschriebenen neurobiologischen Entität) und vertritt die Ansicht, dass neuropsychoanalytische Autoren dazu neigen, den einzigartigen, unhintergehbaren Charakter gelebter Erfahrung und die fundamentalen Unterschiede zwischen der psychoanalytischen Psyche/Geist (die/der ein erfahrendes Subjekt voraussetzt) und dem Gehirn (einer Ansammlung von Neuronen) zunächst anzuerkennen, aber dann zu ignorieren. Der Autor beschreibt eine philosophische Grundlage für die These, dass der Psychoanalyse Gefahren drohen, wenn die wesentlichen

[24]In his critique of Ricoeur, Grünbaum (2004) faults hermeneuticians for confounding Freud's idea of "meaning," which is tied to psychic causation, with a semantic or linguistic notion of "meaning," which refers to signification or representation. One problem with Grünbaum's position, it seems to me, is that psychoanalytic causation is so inextricably tied to personal meaning, a view that is very different from the confounding of a causative idea of meaning with a semantic one.

Unterschiede, die die Neurowissenschaften ihr gegenüber aufweisen, verkannt werden und man die Neurowissenschaft unüberlegt als psychoanalytischen Partner betrachtet, um Fragen zu beantworten, die eigentlich in den Bereich der Sprache und der konzeptuellen Architektur der Psychoanalyse fallen.

Già nel momento stesso in cui tentiamo di comprendere come si relazionino tra loro le due metà che, senza nemmeno essere separate da un trattino, compongono la parola neuropsicoanalisi, la natura inequivocabilmente ibrida di questa disciplina ci invita a riconsiderare criticamente i suoi assunti rispetto alla conoscenza e alla soggettività. Guardando alle differenze tra la mente (fondata nell'esperienza soggettiva) e il cervello (che è invece un'entità neurobiologica descritta oggettivamente), l'autore suggerisce che gli scrittori neuropsicoanalitici sono inclinati in un primo momento a riconoscere ma poi di fatto a ignorare la particolare, irriducibile natura dell'esperienza vissuta, e con essa le fondamentali differenze tra la mente di cui parla la psicoanalisi (che necessita di un soggetto dell'esperienza) e il cervello (che è un aggregato neuronale). L'autore offre con il presente contributo una base filosofica per argomentare che la psicoanalisi corre potenzialmente dei rischi quando le fondamentali differenze che la separano dalle neuroscienze non vengano adeguatamente riconosciute, e anche laddove le neuroscienze vengono di conseguenza improvvidamente utilizzate come "partner psicoanalitico" per rispondere a domande che ha senso porre soltanto all'interno del linguaggio e dell'architettura concettuale della psicoanalisi.

El neuropsicoanálisis como entidad conspicuamente híbrida nos exige examinar críticamente sus supuestos acerca del conocimiento y la subjetividad en cuanto se intenta comprender de qué manera se relacionan entre sí sus mitades sin guion de por medio. El autor analiza las diferencias entre mente (basada en la experiencia subjetiva) y cerebro (entidad neurobiológica descrita de forma objetiva), y sugiere que los escritores de neuropsicoanálisis se inclinan a reconocer, pero luego a pasar por alto, la naturaleza única e irreductible de la experiencia vivida y las diferencias fundamentales entre mente psicoanalítica (que requiere un sujeto experimentador) y cerebro (un agregado neuronal). El autor brinda una base filosófica para argumentar que existen peligros potenciales para el psicoanálisis cuando se obvian las diferencias fundamentales con las neurociencias y se las usa con ligereza como socio psicoanalítico a fin de responder preguntas que pertenecen propiamente al lenguaje y a la arquitectura conceptual del psicoanálisis.

References

Bennett, M. R., and P. M. S. Hacker. 2003. *The Philosophical Foundations of Neuroscience*. Oxford: Blackwell.

Bennett, M. R., and P. M. S. Hacker. 2007. *Neuroscience and Philosophy: Brain, Mind, and Language*. New York: Columbia UP.

Binswanger, L. 1963. *Being in the World: Selected Papers of Ludwig Binswanger*. Translated by J. Needleman. New York: Basic Books.

Blass, R. B., and Z. Carmeli. 2007. "The Case Against Neuropsychoanalysis: On Fallacies Underlying Psychoanalysis' Latest Scientific Trend and its Negative Impact on Psychoanalytic Discourse." *The International Journal of Psycho-Analysis* 88: 19–40.

Blass, R. B., and Z. Carmeli. 2015. "Further Evidence for the Case Against Neuropsychoanalysis: How Yovell, Solms, and Fotopoulou's Response to our Critique Confirms the Irrelevance and Harmfulness to Psychoanalysis of the Contemporary Neuroscientific Trend." *The International Journal of Psycho-Analysis* 96: 1555–1573.

Brothers, L. 2002. "The Trouble with Neurobiological Explanations of the Mind." *Psychoanalytic inquiry* 22: 857–870.

Caws, P. 2003. "Psychoanalysis as the Idiosyncratic Science of the Individual Subject." *Psychoanalytic Psychology* 20: 618–634.

Chalmers, D. J. 1996. *The Conscious Mind: In Search of a Fundamental Theory*. New York: Oxford UP.

Chalmers, D. J. 2004. "How can we Construct a Science of Consciousness?" In *The Cognitive Neurosciences III*, edited by M. Gazzaniga. Cambridge, MA: MIT Press.

Chalmers, D. J. 2010. *The Character of Consciousness*. Oxford: Oxford UP.

Changeux, J.-P., and P. Ricoeur. 2003. *What Makes us Think: A Neuroscientist and a Philosopher Argue About Ethics, Human Nature, and the Brain*. Translated by M. B. DeBevoise. Princeton: Princeton UP.
Chessick, R. D. 2001. "The Contemporary Failure of Nerve and the Crisis in Psychoanalysis." *Journal of the American Academy of Psychoanalysis and Psychodynamic Psychiatry* 29: 659–678.
Cooper, A. M. 1993. "Discussion: On Empirical Research." *Journal of the American Psychoanalytic Association* 41: 381–392.
Damasio, A. 2010. *Self comes to Mind: Constructing the Conscious Brain*. New York: Vintage Books.
Damasio, A. 2012. "Neuroscience and Psychoanalysis: A Natural Alliance." *Psychoanalytic Review* 99 (4): 591–594.
Dennett, D. 1991. *Consciousness Explained*. Boston: Little, Brown.
Dennett, D. 1996. *Kinds of Minds*. New York: Basic Books.
Fonagy, P. 2003. "Genetics, Developmental Psychopathology, and Psychoanalytic Theory: The Case for Ending our (not so) Splendid Isolation." *Psychoanalytic inquiry* 23: 218–247.
Fotopoulou, A. 2012a. "Background, Ethos, and Content." In *From the Couch to the lab: Trends in Psychodynamic Neuroscience*, edited by A. Fotopoulou, D. Pfaff, and M. A. Conway. Oxford: Oxford UP.
Fotopoulou, A. 2012b. "The History and Progress of Neuropsychoanalysis." In *From the Couch to the lab: Trends in Psychodynamic Neuroscience*, edited by A. Fotopoulou, D. Pfaff, and M. A. Conway. Oxford: Oxford UP.
Fotopoulou, A. 2012c. "Towards a Psychodynamic Neuroscience." In *From the Couch to the lab: Trends in Psychodynamic Neuroscience*, edited by A. Fotopoulou, D. Pfaff, and M. A. Conway. Oxford: Oxford UP.
Freud S (1900). The interpretation of dreams. SE 4-5.
Frie, R. 2004. "Formulating Unconscious Experience: From Freud to Binswanger and Sullivan." In *Psychoanalysis at the Limit: Epistemology, Mind, and the Question of Science*, edited by J. Mills. Albany, NY: State University of New York Press.
Gabbard, G. 2000a. "The Integration of Neuroscientific Thought in the Realm of a Biologically Informed Psychodynamic Model of Personality Disorder." In *Changing Ideas in a Changing World: The Revolution in Psychoanalysis: Essays in Honour of Arnold Cooper*, edited by J. Sandler, R. Michels, and P. Fonagy, 155–162. London: Karnac Books.
Gabbard, G. 2000b. "What can Neuroscience Teach us About Transference?" *Canadian Journal of Psychoanalysis* 9: 1–18.
Gadamer, H.-G. 1960. *Truth and Method*. 2nd rev. Translated by J. Weinsheimer and D. G. Marshall. New York: Crossroad, 1989.
Gadamer, H.-G. 1981. "Hermeneutics as Practical Philosophy." In *Reason in the age of Science*, 88–112. Translated by F. G. Lawrence. Cambridge, MA: MIT Press.
Ginot, E. 2015. *The Neuropsychology of the Unconscious: Integrating Brain and Mind in Psychotherapy*. New York: Norton.
Green, A. 1999. "Consilience and Rigor: Commentary by Andre Green." *Neuropsychoanalysis* 1: 40–44.
Green, A. 2000. "Commentary." *Journal of the American Psychoanalytic Association* 48: 57–66.
Green, A. 2001. "Advice to Psychoanalysts: "Cognitive Psychology is Good for you": Commentary." *Neuropsychoanalysis* 3: 16–19.
Green, A. 2002. "The Crisis in Psychoanalytic Understanding." *Fort da* 8: 58–71.
Green, V. 2003. *Emotional Development in Psychoanalysis, Attachment Theory and Neuroscience: Creating Connections*. New York: Brunner-Routledge.
Green, A. 2005a. *Key Ideas for a Contemporary Psychoanalysis: Misrecognition and Recognition of the Unconscious*. Translated by A. Weller. London: Routledge.
Green, A. 2005b. "The Illusion of Common Ground and Mythical Pluralism." *The International Journal of Psycho-Analysis* 86: 627–632.
Green, A. 2008. "Freud's Concept of Temporality: Differences with Current Ideas." *The International Journal of Psycho-Analysis* 89: 1029–1039.
Grünbaum, A. 2004. "The Hermeneutic Versus the Scientific Conception of Psychoanalysis." In *Psychoanalysis at the Limit: Epistemology, Mind, and the Question of Science*, edited by J. Mills. Albany, NY: State University of New York Press.

Hanly, C. 1999. “On Subjectivity and Objectivity in Psychoanalysis.” *Journal of the American Psychoanalytic Association* 47: 427–444.
Hanly, C. 2006. “Pragmatism, Tradition, and Truth in Psychoanalysis.” *American Imago* 63: 261–282.
Hanly, C. 2014a. “The Interplay of Deductive and Inductive Reasoning in Psychoanalytic Theorizing.” *The Psychoanalytic Quarterly* 83 (4): 897–915.
Hanly, C. 2014b. “Skeptical Reflection on Subjectivist Epistemologies.” *Psychoanal Quarterly* 83 (4): 949–968.
Hanly, C., and M. A. Fitzpatrick Hanly. 2001. “Critical Realism: Distinguishing the Psychological Subjectivity of the Analyst From Epistemological Subjectivism.” *Journal of the American Psychoanalytic Association* 49: 515–532.
Hass, L. 2008. *Merleau-Ponty’s Philosophy*. Bloomington, IN: Indiana UP.
Hoffman, I. Z. 2009. “Doublethinking our way to “Scientific” Legitimacy: The Desiccation of Human Experience.” *Journal of the American Psychoanalytic Association* 57: 1043–1069.
Kandel, E. R. 1998. “A new Intellectual Framework for Psychiatry.” *American Journal of Psychiatry* 155: 457–469.
Kandel, E. R. 2005. “Biology and the Future of Psychoanalysis: A new Intellectual Framework for Psychiatry Revisited.” In *Psychiatry, Psychoanalysis, and the new Biology of Mind*. Washington, DC: American Psychiatric Publishing. Reprinted from *Am J Psychiatry*, *156:505-524*, 1999.
Kaplan-Solms, K., and M. Solms. 2000. *Clinical Studies in Neuro-Psychoanalysis: Introduction to a Depth Neuropsychology*. London: Karnac Books.
Karlsson, G. 2010. *Psychoanalysis in a new Light*. Cambridge: Cambridge UP.
Loewald, H. 1971. “On Motivation and Instinct Theory.” In *Papers on Psychoanalysis*. New Haven, CT: Yale UP. 1980.
Lothane, Z. 1998. “Freud’s 1895 *Project*: From Mind to Brain and Back Again.” In *Neuroscience of the Mind on the Centennial of Freud’s Project for a Scientific Psychology*, edited by R. M. Bilder and F. F. LeFever. New York: The New York Academy of Sciences.
Mayes, L. C. 2005. “Interface Between Psychoanalytic Developmental Theory and Other Disciplines.” In *Textbook of Psychoanalysis*, edited by E. S. Person, A. M. Cooper, and G. O. Gabbard. Washington, D.C.: American Psychiatric Publishing, Inc.
Merleau-Ponty, M. 1945. *Phenomenology of Perception*. Translated by D. A. Landes. New York: Routledge. 2012.
Merleau-Ponty, M. (1960). Phenomenology and psychoanalysis: Preface to Hesnard’s *L’Oeuvre de Freud*. Translated by A. Fisher. *Review of Existential Psychology and Psychiatry: Merleau-Ponty and Psychology* 18, vols. 1,2,3, 1982. Reprinted *Merleau-Ponty and Psychology*, Hoeller K, editor, p. 67-72. Atlantic Highlands, NJ: Humanities Press International, 1993.
Merleau-Ponty, M. 1964. *The Visible and the Invisible*. Edited by C. Lefort; Translated by A. Lingis. Evanston, IL: Northwestern UP, 1968.
Mills, J. 2012. *Conundrums: A Critique of Contemporary Psychoanalysis*. New York: Routledge.
Modell, A. H. 1993. *The Private Self*. Cambridge, MA: Harvard UP.
Modell, A. H. 2003. *Imagination and the Meaningful Brain*. Cambridge, MA: MIT Press.
Modell, A. H. 2012a. “Psychoanalysis, Neuroscience, and the Unconscious Self.” *Psychoanalytic Review* 99 (4): 473–483.
Modell, A. H. 2012b. “Science Confirms Psychoanalytic Theory: Commentary on Howard Shevrin’s Paper.” *Psychoanalytic Review* 99 (4): 511–513.
Modell, A. H. 2013a. “Some Comments on the Freudian Unconscious.” *Psychoanalytic Review* 100 (4): 535–541.
Modell, A. H. 2013b. “Metaphor, Meaning, and the Mind.” In *Metaphor and Fields: Common Ground, Common Language, and the Future of Psychoanalysis*, edited by S. M. Katz, 59–66. Psychoanalytic Inquiry Book Series. New York: Routledge.
Morag, T. 2016. *Emotion, Imagination, and the Limits of Reason*. New York: Routledge.
Nagel, T. 1986. *The View From Nowhere*. New York: Oxford UP.
Nagel, T. 1995a. “Consciousness Dissolved.” In *Other Minds: Critical Essays 1969-1994*. New York: Oxford UP.

Nagel, T. 1995b. “Searle: Why we are not Computers.” In *Other Minds: Critical Essays 1969-1994*. New York: Oxford UP.

Nagel, T. 2012. *Mind & cosmos: Why the Materialist neo-Darwinian Conception of Nature is Almost Certainly False*. New York: Oxford Press.

Olds, D. 2012. “What if Freud Hadn’t: Commentary on Arnold Modell’s Paper.” *Psychoanalytic Review* 99 (4): 485–489.

Olds, D., and A. M. Cooper. 1997. “Dialogue with Other Sciences: Opportunities for Mutual Gain.” *The International Journal of Psycho-Analysis* 78: 219–225.

Opatow, B. 1999. “Affect and the Integration Problem of Mind and Brain.” *Neuropsychoanalysis* 1: 97–110.

Orange, D. M. 2001. “From Cartesian Minds to Experiential Worlds in Psychoanalysis.” *Psychoanal Psychology* 18: 287–302.

Orange, D. M. 2003. “Antidotes and Alternatives: Perspectival Realism and the new Reductionisms.” *Psychoanal Psychology* 20: 472–486.

Pally, R. 2000. *The Mind-Brain Relationship*. International Journal of Psychoanalysis Key Papers Series. London: Karnac Books.

Pulver, S. E. 2003. “On the Astonishing Clinical Irrelevance of Neuroscience.” *Journal of the American Psychoanalytic Association* 51: 755–772.

Ricoeur, P. 1970. *Freud and Philosophy: An Essay on Interpretation*. Translated by D. Savage. New Haven: Yale UP.

Ricoeur, P. 1973. “The Hermeneutical Function of Distanciation.” In *From Text to Action: Essays in Hermeneutics, II*, 75–88. Translated by K. Blamey and J. B. Thompson. Evanston, IL: Northwestern UP, 1991.

Rothman J. (2017). “A Science of the Soul: A Philosopher’s Quest to Understand the Making of the Mind.” *The New Yorker*, 27 March, 2017, p. 46-55.

Sacks, O. W. 1998. “Sigmund Freud: The Other Road.” In *Freud and the Neurosciences: From Brain Research to the Unconscious*, edited by G. Guttmann and I. Scholz-Strasser. Vienna: Verlag der Osterreichischen Akademie der Wissenschaften.

Safran, J. D. 2012. “Doublethinking or Dialectical Thinking: A Critical Appreciation of Hoffman’s “Doublethinking” Critique.” *Psychoanalytic Dialog* 22: 710–720.

Sass, L. A. 2015. “Lacan: The Mind of the Modernist.” *Continental Philosophy Review* 48: 409–443.

Scarfone, D. 2012. “ On: Minding our Metaphysics [Letter].” *The International Journal of Psycho-Analysis* 93: 1286–1288.

Schore, A. N. 1994. *Affect Regulation and the Origin of the Self: The Neurobiology of Emotional Development*. Mahwah, NJ: Erlbaum.

Schore, A. N. 2003. “The Human Unconscious: The Development of the Right Brain and its Role in Early Emotional Life.” In *Emotional Development in Psychoanalysis, Attachment Theory and Neuroscience*, edited by V. Green. New York: Brunner-Routledge.

Schore, A. N. 2012. *The Science of the art of Psychotherapy*. New York: Norton.

Shevrin, H. 1999a. “Commentary by Howard Shevrin.” *Neuropsychoanalysis* 1: 55–60.

Shevrin, H. 1999b. “Jaak Panksepp’s Response: Commentary by Howard Shevrin.” *Neuropsychoanalysis* 1: 247–250.

Shevrin, H. 2012. “A Contribution Toward a Science of Psychoanalysis.” *Psychoanalytic Review* 99: 491–509.

Solms, M. 1998. “Before and After Freud’s *Project*.” In *Neuroscience of the Mind on the Centennial of Freud’s Project for a Scientific Psychology*, edited by R. M. Bilder and F. F. LeFever. New York: The New York Academy of Sciences.

Solms, M. 2000. "Foreward." In *The Mind-Brain Relationship*, by R. Pally. International Journal of Psychoanalysis Key Paper Series. London: Karnac Books.

Solms, M. 2003. “Do Unconscious Fantasies Really Exist?” In *Unconscious Fantasy*, edited by R. Steiner. London: Karnac.

Solms, M., and O. Turnbull. 2002. *The Brain and the Inner World: An introduction to the Neuroscience of Subjective Experience*. New York: Other Press.

Thompson, M. G. 2012. "The Role of Being and Experience in Freud's Ontology of the Unconscious." In *Psychoanalysis at the Limit: Epistemology, Mind, and the Question of Science*, edited by J. Mills. Albany, NY: State University of New York Press. Originally published, *Contemporary Psychoanalysis* 37: 571-612 (2001).

Turnbull, O., and M. Solms. 2003. "Memory, Amnesia and Intuition: A Neuro-Psychoanalytic Perspective." In *Emotional Development in Psychoanalysis, Attachment Theory and Neuroscience: Creating Connections*, edited by V. Green. New York: Brunner-Routledge.

Varela, F. J., and J. Shear. 1999. *The View From Within: First-Person Approaches to the Study of Consciousness*. Exeter, UK: Imprint Academic.

Westen, D., and G. Gabbard. 2002a. "Developments in Cognitive Neuroscience: I. Conflict, Compromise, and Connectionism." *Journal of the American Psychoanalytic Association* 50: 53–98.

Westen, D., and G. Gabbard. 2002b. "Developments in Cognitive Neuroscience: II. Implications for Theories of Transference." *Journal of the American Psychoanalytic Association* 50: 99–134.

Westphal, M. 2011. "The Dialectic of Belonging and Distanciation in Gadamer and Ricoeur." In *Gadamer and Ricoeur: Critical Horizons for Contemporary Hermeneutics*, edited by F. J. Mootz, III and G. H. Taylor, 43–62. London: Bloomsbury.

Yovell, Y., M. Solms, and A. Fotopoulou. 2015. "The Case for Neuropsychoanalysis: Why a Dialogue with Neuroscience is Necessary but not Sufficient for Psychoanalysis." *The International Journal of Psychoanalysis* 96: 1515–1553.

Zellner, M. R. 2012. "Toward a Materialist Metapsychology: Major Operating Principles of the Brain Provide a Blueprint for a Fundamentally Psychodynamic Infrastructure." *Psychoanalytic Review* 99 (4): 563–588.